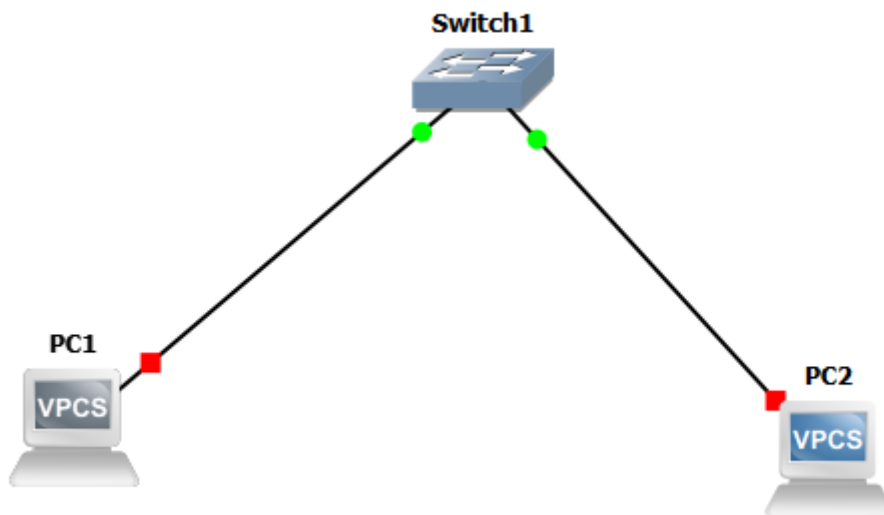


1) Установить и настроить эмулятор GNS3

2) Создать простейшую сеть, состоящую из 1 коммутатора и 2 компьютеров, назначить им произвольные ip адреса из одной сети



PC1> ip 192.168.0.1/24

Checking for duplicate address...

PC1 : 192.168.0.1 255.255.255.0

PC2> ip 192.168.0.2/24

Checking for duplicate address...

PC2 : 192.168.0.2 255.255.255.0

3) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя ip адрес второго компьютера

PC1> ping 192.168.0.2

84 bytes from 192.168.0.2 icmp_seq=1 ttl=64 time=0.310 ms

84 bytes from 192.168.0.2 icmp_seq=2 ttl=64 time=0.294 ms

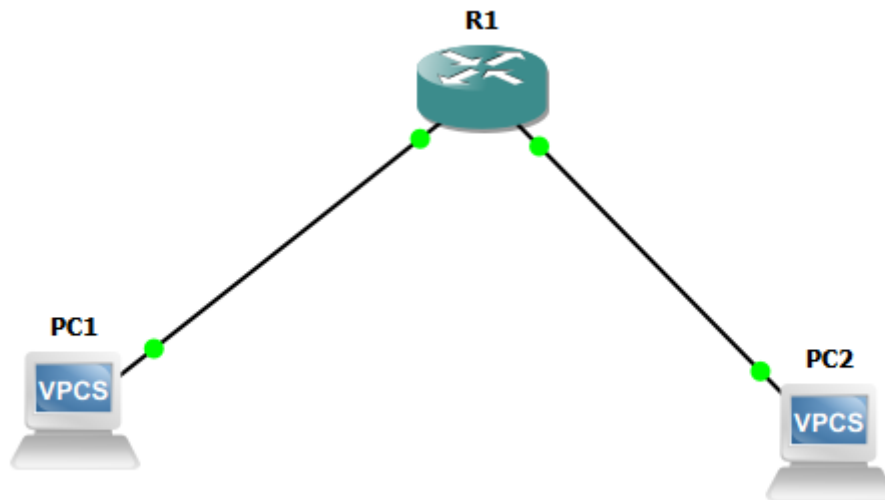
84 bytes from 192.168.0.2 icmp_seq=3 ttl=64 time=0.265 ms

84 bytes from 192.168.0.2 icmp_seq=4 ttl=64 time=0.263 ms

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.0.2? Tell 192.168.0.1
2	0.000154	Private_66:68:01	Private 66:68:00	ARP	64	192.168.0.2 is at 00:50:79:66:68:01

```
> Frame 1: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
▼ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  > Destination: Broadcast (ff:ff:ff:ff:ff:ff)
  > Source: Private_66:68:00 (00:50:79:66:68:00)
    Type: ARP (0x0806)
    [Stream index: 0]
    Padding: 00000000000000000000000000000000
    Frame check sequence: 0x00000000 [unverified]
    [FCS Status: Unverified]
▼ Address Resolution Protocol (request)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Sender IP address: 192.168.0.1
  Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
  Target IP address: 192.168.0.2
```

```
> Frame 2: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
▼ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)
    > Destination: Private_66:68:00 (00:50:79:66:68:00)
    > Source: Private_66:68:01 (00:50:79:66:68:01)
        Type: ARP (0x0806)
        [Stream index: 1]
        Padding: 0000000000000000000000000000000000
        Frame check sequence: 0x00000000 [unverified]
        [FCS Status: Unverified]
▼ Address Resolution Protocol (reply)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
    Sender IP address: 192.168.0.2
    Target MAC address: Private_66:68:00 (00:50:79:66:68:00)
    Target IP address: 192.168.0.1
```

```
PC1> ip 192.168.0.1 24 192.168.0.2
```

```
Checking for duplicate address...
```

```
PC1 : 192.168.0.1 255.255.255.0 gateway 192.168.0.2
```

```
PC2> ip 192.168.1.1 24 192.168.1.2
```

```
Checking for duplicate address...
```

```
PC2 : 192.168.1.1 255.255.255.0 gateway 192.168.1.2
```

```
R1#enable
```

```
R1#conf t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
R1(config)#int fa0/0
```

```
R1(config-if)#ip address 192.168.0.2 255.255.255.0
```

```
R1(config-if)#no sh
```

```
R1(config-if)#exit
```

```
R1(config)#
```

```
*Mar 1 00:05:41.007: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
```

```
*Mar 1 00:05:42.007: %LINEPROTO-5-UPDOWN: Line protocol on Interface  
FastEthernet0/0, changed state to up
```

```
R1(config)#int fa1/0
```

```
R1(config-if)#ip address 192.168.1.2 255.255.255.0
```

```
R1(config-if)#no sh
```

```
R1(config-if)#exit
```

```
R1(config)#
```

```
*Mar 1 00:06:31.107: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
```

```
*Mar 1 00:06:32.107: %LINEPROTO-5-UPDOWN: Line protocol on Interface  
FastEthernet1/0, changed state to up
```

```
R1(config)#exit
```

```
R1#
```

```
*Mar 1 00:06:34.059: %SYS-5-CONFIG_I: Configured from console by console
```

6) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя IP-адрес второго компьютера

```
PC1> ping 192.168.1.1
```

```
192.168.1.1 icmp_seq=1 timeout
```

```
84 bytes from 192.168.1.1 icmp_seq=2 ttl=63 time=20.469 ms
```

```
84 bytes from 192.168.1.1 icmp_seq=3 ttl=63 time=15.365 ms
```

```
84 bytes from 192.168.1.1 icmp_seq=4 ttl=63 time=15.657 ms
```

```
84 bytes from 192.168.1.1 icmp_seq=5 ttl=63 time=15.802 ms
```

7) Перехватить трафик протокола arp и icmp на всех линках(nb!), задокументировать и проанализировать заголовки пакетов в программе Wireshark, для фильтрации трафика, относящегося к указанному протоколу использовать фильтры Wireshark

Между PC1 и коммутатором:

No.	Time	Source	Destination	Protocol	Length	Info
6	30.641958	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.0.2? Tell 192.168.0.1
7	30.647688	cc:01:57:77:00:00	Private_66:68:00	ARP	60	192.168.0.2 is at cc:01:57:77:00:00
8	30.648235	192.168.0.1	192.168.1.1	ICMP	98	Echo (ping) request id=0xf740, seq=1/256, ttl=64 (no response found!)
9	32.648544	192.168.0.1	192.168.1.1	ICMP	98	Echo (ping) request id=0xf940, seq=2/512, ttl=64 (reply in 10)
10	32.668936	192.168.1.1	192.168.0.1	ICMP	98	Echo (ping) reply id=0xf940, seq=2/512, ttl=63 (request in 9)
11	33.669862	192.168.0.1	192.168.1.1	ICMP	98	Echo (ping) request id=0xfa40, seq=3/768, ttl=64 (reply in 12)
12	33.685127	192.168.1.1	192.168.0.1	ICMP	98	Echo (ping) reply id=0xfa40, seq=3/768, ttl=63 (request in 11)
13	34.685831	192.168.0.1	192.168.1.1	ICMP	98	Echo (ping) request id=0xfb40, seq=4/1024, ttl=64 (reply in 14)
14	34.701400	192.168.1.1	192.168.0.1	ICMP	98	Echo (ping) reply id=0xfb40, seq=4/1024, ttl=63 (request in 13)
15	35.701728	192.168.0.1	192.168.1.1	ICMP	98	Echo (ping) request id=0xfc40, seq=5/1280, ttl=64 (reply in 16)
16	35.717440	192.168.1.1	192.168.0.1	ICMP	98	Echo (ping) reply id=0xfc40, seq=5/1280, ttl=63 (request in 15)

```

> Frame 6: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
▼ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  > Destination: Broadcast (ff:ff:ff:ff:ff:ff)
  > Source: Private_66:68:00 (00:50:79:66:68:00)
    Type: ARP (0x0806)
    [Stream index: 2]
    Padding: 0000000000000000000000000000000000000000000000000000000000000000
    Frame check sequence: 0x00000000 [unverified]
    [FCS Status: Unverified]
▼ Address Resolution Protocol (request)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Sender IP address: 192.168.0.1
  Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
  Target IP address: 192.168.0.2

> Frame 7: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
▼ Ethernet II, Src: cc:01:57:77:00:00 (cc:01:57:77:00:00), Dst: Private_66:68:00 (00:50:79:66:68:00)
  > Destination: Private_66:68:00 (00:50:79:66:68:00)
  > Source: cc:01:57:77:00:00 (cc:01:57:77:00:00)
    Type: ARP (0x0806)
    [Stream index: 3]
    Padding: 0000000000000000000000000000000000000000000000000000000000000000
▼ Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: cc:01:57:77:00:00 (cc:01:57:77:00:00)
  Sender IP address: 192.168.0.2
  Target MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Target IP address: 192.168.0.1

```

Был ARP запрос на Broadcast и ARP ответ для определения MAC адреса.

```

> Frame 8: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
▼ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: cc:01:57:77:00:00 (cc:01:57:77:00:00)
  > Destination: cc:01:57:77:00:00 (cc:01:57:77:00:00)
  > Source: Private_66:68:00 (00:50:79:66:68:00)
    Type: IPv4 (0x0800)
    [Stream index: 3]
▼ Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.1.1
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 84
    Identification: 0x40f7 (16631)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: ICMP (1)
    Header Checksum: 0xb75f [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.1
    Destination Address: 192.168.1.1
    [Stream index: 0]
> Internet Control Message Protocol

```

Первый ICMP запрос не получил ответа.

```
> Frame 9: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
▼ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: cc:01:57:77:00:00 (cc:01:57:77:00:00)
  > Destination: cc:01:57:77:00:00 (cc:01:57:77:00:00)
  > Source: Private_66:68:00 (00:50:79:66:68:00)
  Type: IPv4 (0x0800)
  [Stream index: 3]
▼ Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.1.1
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 84
  Identification: 0x40f8 (16632)
  > 000. .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 64
  Protocol: ICMP (1)
  Header Checksum: 0xb75e [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.0.1
  Destination Address: 192.168.1.1
  [Stream index: 0]
> Internet Control Message Protocol

> Frame 10: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
▼ Ethernet II, Src: cc:01:57:77:00:00 (cc:01:57:77:00:00), Dst: Private_66:68:00 (00:50:79:66:68:00)
  > Destination: Private_66:68:00 (00:50:79:66:68:00)
  > Source: cc:01:57:77:00:00 (cc:01:57:77:00:00)
  Type: IPv4 (0x0800)
  [Stream index: 3]
▼ Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.0.1
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 84
  Identification: 0x40f8 (16632)
  > 000. .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 63
  Protocol: ICMP (1)
  Header Checksum: 0xb85e [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.1.1
  Destination Address: 192.168.0.1
  [Stream index: 0]
> Internet Control Message Protocol
```

Следующие запросы получили ответы. В заголовках используется MAC второго компьютера, определённый через ARP и IP обоих компьютеров.

Между PC2 и коммутатором:

No.	Time	Source	Destination	Protocol	Length	Info
4	20.629224	cc:01:57:77:00:10	Broadcast	ARP	60	Who has 192.168.1.1? Tell 192.168.1.2
5	20.629417	Private_66:68:01	cc:01:57:77:00:10	ARP	60	192.168.1.1 is at 00:50:79:66:68:01
6	22.630300	192.168.0.1	192.168.1.1	ICMP	98	Echo (ping) request id=0xf940, seq=2/512, ttl=63 (reply in 7)
7	22.630439	192.168.1.1	192.168.0.1	ICMP	98	Echo (ping) reply id=0xf940, seq=2/512, ttl=64 (request in 6)
8	23.646526	192.168.0.1	192.168.1.1	ICMP	98	Echo (ping) request id=0xfa40, seq=3/768, ttl=63 (reply in 9)
9	23.646683	192.168.1.1	192.168.0.1	ICMP	98	Echo (ping) reply id=0xfa40, seq=3/768, ttl=64 (request in 8)
10	24.662779	192.168.0.1	192.168.1.1	ICMP	98	Echo (ping) request id=0xfb40, seq=4/1024, ttl=63 (reply in 11)
11	24.662935	192.168.1.1	192.168.0.1	ICMP	98	Echo (ping) reply id=0xfb40, seq=4/1024, ttl=64 (request in 10)
12	25.678771	192.168.0.1	192.168.1.1	ICMP	98	Echo (ping) request id=0xfc40, seq=5/1280, ttl=63 (reply in 13)
13	25.678923	192.168.1.1	192.168.0.1	ICMP	98	Echo (ping) reply id=0xfc40, seq=5/1280, ttl=64 (request in 12)


```
> Frame 4: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
▼ Ethernet II, Src: cc:01:57:77:00:10 (cc:01:57:77:00:10), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
    > Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    > Source: cc:01:57:77:00:10 (cc:01:57:77:00:10)
        Type: ARP (0x0806)
        [Stream index: 1]
        Padding: 0000000000000000000000000000000000000000
▼ Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: cc:01:57:77:00:10 (cc:01:57:77:00:10)
    Sender IP address: 192.168.1.2
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 192.168.1.1

> Frame 5: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
▼ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: cc:01:57:77:00:10 (cc:01:57:77:00:10)
    > Destination: cc:01:57:77:00:10 (cc:01:57:77:00:10)
    > Source: Private_66:68:01 (00:50:79:66:68:01)
        Type: ARP (0x0806)
        [Stream index: 2]
        Padding: 0000000000000000000000000000000000000000
▼ Address Resolution Protocol (reply)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
    Sender IP address: 192.168.1.1
    Target MAC address: cc:01:57:77:00:10 (cc:01:57:77:00:10)
    Target IP address: 192.168.1.2
```

PC2 получает широковещательный ARP-запрос и отвечает на него своим MAC.

Один из ICMP-запросов не дошёл.

```
> Frame 6: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
Ethernet II, Src: cc:01:57:77:00:10 (cc:01:57:77:00:10), Dst: Private_66:68:01 (00:50:79:66:68:01)
  Destination: Private_66:68:01 (00:50:79:66:68:01)
  Source: cc:01:57:77:00:10 (cc:01:57:77:00:10)
  Type: IPv4 (0x0800)
  [Stream index: 2]
Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.1.1
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 84
  Identification: 0x40f8 (16632)
  > 000. .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 63
  Protocol: ICMP (1)
  Header Checksum: 0xb85e [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.0.1
  Destination Address: 192.168.1.1
  [Stream index: 0]
> Internet Control Message Protocol
```



```

> Frame 7: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
▼ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: cc:01:57:77:00:10 (cc:01:57:77:00:10)
  > Destination: cc:01:57:77:00:10 (cc:01:57:77:00:10)
  > Source: Private_66:68:01 (00:50:79:66:68:01)
  Type: IPv4 (0x0800)
  [Stream index: 2]
▼ Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.0.1
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 84
  Identification: 0x40f8 (16632)
  > 000. .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 64
  Protocol: ICMP (1)
  Header Checksum: 0xb75e [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.1.1
  Destination Address: 192.168.0.1
  [Stream index: 0]
> Internet Control Message Protocol

```

Остальные получены и отвечены с использованием MAC- и IP-адресов компьютеров